

**2** A satellite is in geostationary orbit around the Earth. The orbit is circular.

**(a) (i)** State what is meant by a geostationary orbit.

.....  
.....  
..... [1]

**(ii)** State **two** properties of the orbit.

1 .....  
.....  
2 .....  
..... [2]

**(b)** The orbit has radius  $R$  and period  $T$ .

**(i)** Determine an expression, in terms of  $R$  and  $T$ , for the speed  $v$  of the satellite in its orbit.

$v =$  ..... [1]

**(ii)** Use Newton's law of gravitation to show that the mass  $M$  of the Earth is given by

$$M = \frac{4\pi^2 R^3}{GT^2}$$

where  $G$  is the gravitational constant.

[2]

- (c) The Earth may be considered to be a sphere of radius 6370 km.  
Geostationary satellites orbit at a height of 35 900 km above the surface of the Earth.

Determine a value, to three significant figures, for the mean density of the Earth.

density = .....  $\text{kg m}^{-3}$  [4]

[Total: 10]